

SERIES A · 2026



VEPOLOXAMER · A PHARMACEUTICAL GRADE, PURIFIED FORM OF POLOXAMER 188

A cell membrane-repair platform opportunity

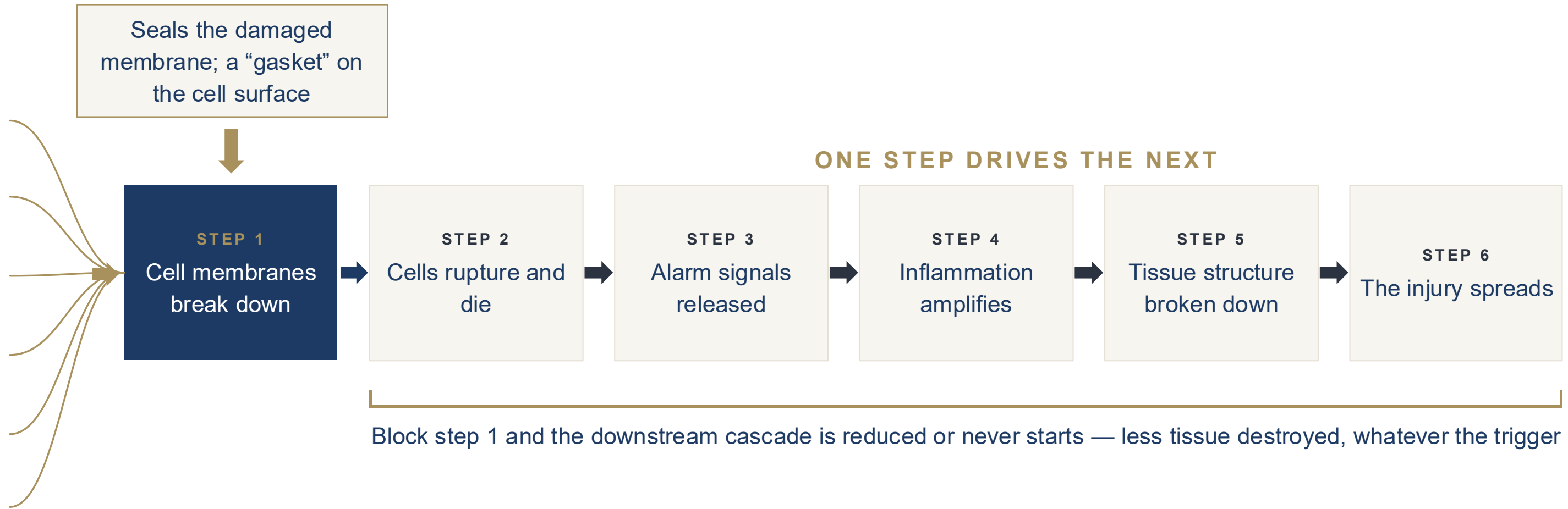
THE PLATFORM

When tissue and cells are injured, it is not the initial injury that does all the damage—it is a cascade of cellular injury after the fact.

OUR DRUG: VEPLOXAMER

INJURY TRIGGERS

- Radiation therapy
LEAD PROGRAM
- Chemotherapy
- Heart attack / stroke
(blood flow returns)
- Blood loss and resuscitation
- Physical trauma
- Thermal burn
SECOND PROGRAM



Because the target is a physical state of the cell membrane, the same mechanism applies wherever membranes are being damaged. That is what makes this a platform, not a single-indication drug.

One mechanism, two treatment opportunities; dosed before the injury, or in the hours after it

DOSED BEFORE

Dosing is elective and scheduled, so the drug is already circulating when the injury occurs. Trials are smaller, faster and cheaper.

POTENTIAL INDICATIONS

Radiation-induced severe oral mucositis — lead

Radiation proctitis, esophagitis, dermatitis, pneumonitis

Radiological and nuclear countermeasures (BARDA, DoD)

Planned ischemia in surgery and organ preservation

DOSED SOON AFTER

Dosed in the hours after an unplanned injury. Hundreds of thousands of patients a year, almost no approved options — the larger prize.

POTENTIAL INDICATIONS

Acute burns — second readout

Traumatic brain injury

Ischemic stroke as a tPA and/or thrombectomy adjunct; cardiac protection

Hemorrhagic shock and resuscitation

This round funds one trial in each franchise — a Phase 1b/2a in radiation-induced severe oral mucositis and a single-site Phase 1 in acute burns — so the company holds human data on both sides of the platform by 2029.

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Lead indications and strategy

Lead indication: a **first-in-class, de-risked** therapy for radiation-induced severe oral mucositis (RI-SOM)



High Unmet Need: ~90K Radiation Therapy patients/yr across the US, EU & UK face this debilitating toxicity with no FDA-approved preventive; >\$1B addressable market



First-in-Class Mechanism: PURIFIED FORM OF POLOXAMER 188 physically reseals damaged cell membranes, mitigating the injury cascade, without preventing the effect of radiation therapy on cancer cells



Safety De-Risked in Other Indications: ~2,300 human exposures in prior sickle-cell (Vepoloxamer) and cardiac (unpurified P188) trials; Vepoloxamer AEs comparable to placebo, no drug-related deaths. Not yet studied clinically in oral mucositis.



Manufacturing In Hand: ~140 kg of GMP drug substance on hand; enough to supply the planned proof-of-concept trial.



Regulatory Head Start: Complete IND package from prior Phase 3 program.



Strong IP & Exclusivity: Composition-of-matter to 2035 (US + ex-US); Method-of-Use (2041+) granted in Japan, under appeal in US/EU; plus NCE market exclusivity.



Capital-Efficient Structure: Clinical proof-of-concept on oral mucositis achievable by 2029 with ~\$30M total; a \$15M seed through first clinical data, then ~\$15M to complete Phase 1b/2a PoC.

“Pain lengthens time.”

A patient's account of severe oral mucositis, published in the *New England Journal of Medicine*.

This is the toxicity Vepoloxamer is designed to prevent — for the 60–70% of head & neck radiation patients who develop it, and for whom no FDA-approved preventive exists.

STOMATITIS

A fish hook lodges in my throat.

Spittle, kindergarten paste, thickens everything – even vision.

Mouth, pocked with sores & blisters, swollen ulcerated tongue.

Topside, sandpapered with number 7 coarsest grade.

Taste buds, saliva glands, seared.

Cool water, corrosive acid now.

The tongue rests; teeth become enemies.

Coiled steel razored wire atop dentate prison walls.

Only moans escape my lips. I cannot eat or speak.

Inside a howl festers.

Pain lengthens time.

— Anita Hart Balter

A devastating, untreated toxicity in a large, well-defined population

Healthy cells of the mouth, while not the target of the radiation, are also damaged during treatment

IMPACT OF SEVERE ORAL MUCOSITIS

- **A majority of patients** require opioids, need a feeding tube or a pureed diet by the end of treatment, and report difficulty eating and swallowing.
- **More than a third** of patients are hospitalized during treatment, with a mean stay of **nearly 5 days**.
- Treatment breaks let tumors regrow **about 3 times faster**, and each missed day of radiation cuts tumor control by about **1%**.



~90K

new US, EU, and UK head & neck cancer patients/yr requiring radiation

60% - 70%

develop severe Grade 3+ mucositis (98.6% develop some OM)

\$1B+

global addressable market (US + EU + UK)

0

FDA-approved drugs to prevent radiation-induced SOM

Vepoloxamer: novel mechanism of action for Radiation-Induced Severe Oral Mucositis, minimizing damage to healthy cells without protecting the tumor



A non-ionic triblock copolymer. Vepoloxamer integrates into damaged cell membranes, acting upstream of the oxidative cascade that drives mucosal injury.



Regulatory head start

A complete filed IND (EPIC Phase 3 in sickle cell); ~75% of that language is reusable for the oral mucositis IND. Failure in sickle cell was wrong indication, not a molecule failure*.



Safety de-risked in other indications

~2,300 human exposures in prior trials in other indications (sickle cell (Vepoloxamer), and cardiac (unpurified P188)) with no drug-related deaths, Vepoloxamer AEs comparable to placebo. Not yet studied clinically in oral mucositis.



Manufacturing in hand

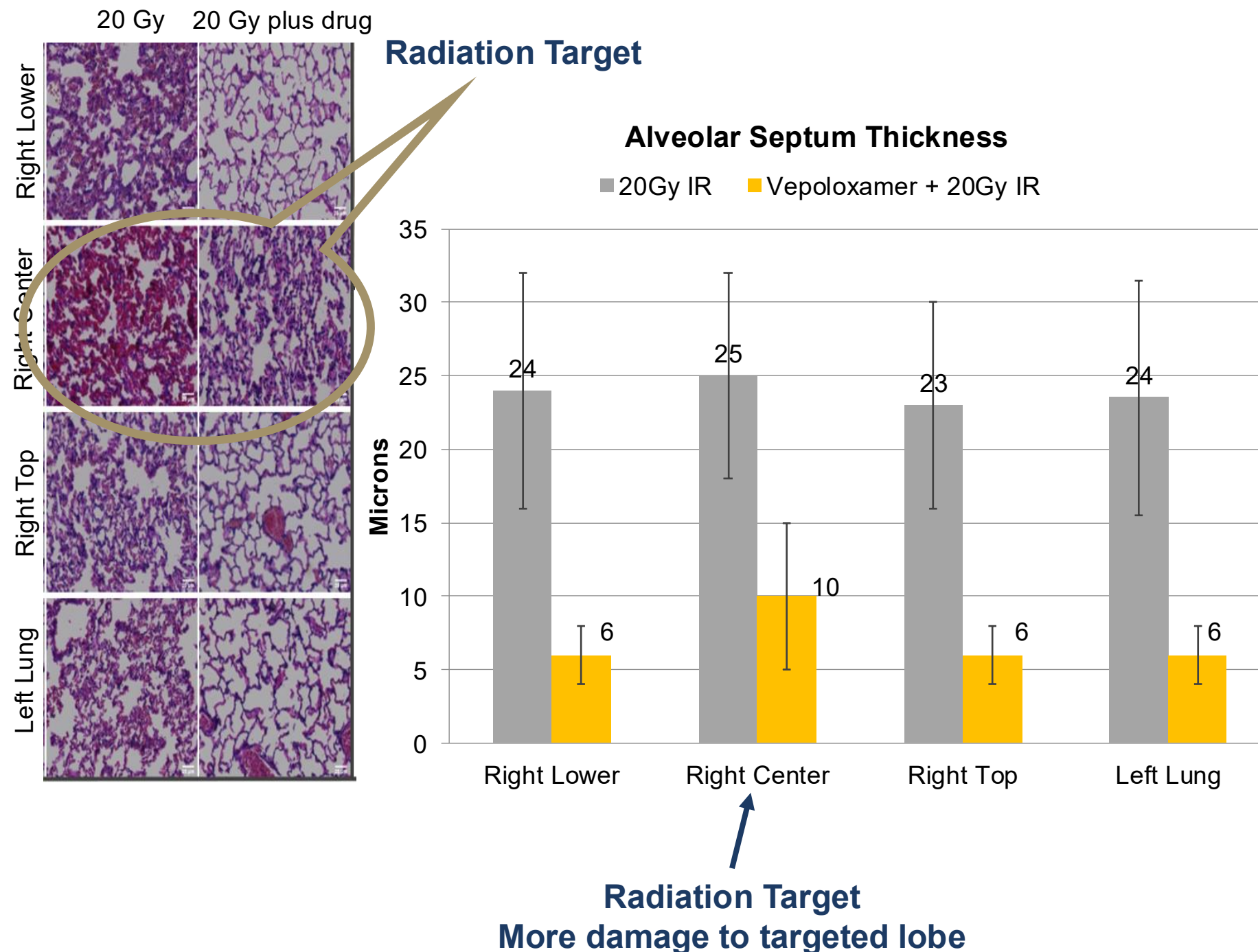
~140 kg of GMP API on hand (enough for initial trials) and a documented, transferable manufacturing process ready for CDMO re-establishment.



Protected value

Composition-of-matter patent to 2035, an exclusively-licensed method-of-use application (approved in Japan), and 5-year NCE exclusivity at launch.

Vepoloxamer reduces damage to non-targeted tissue in animal model exposed to ionizing radiation



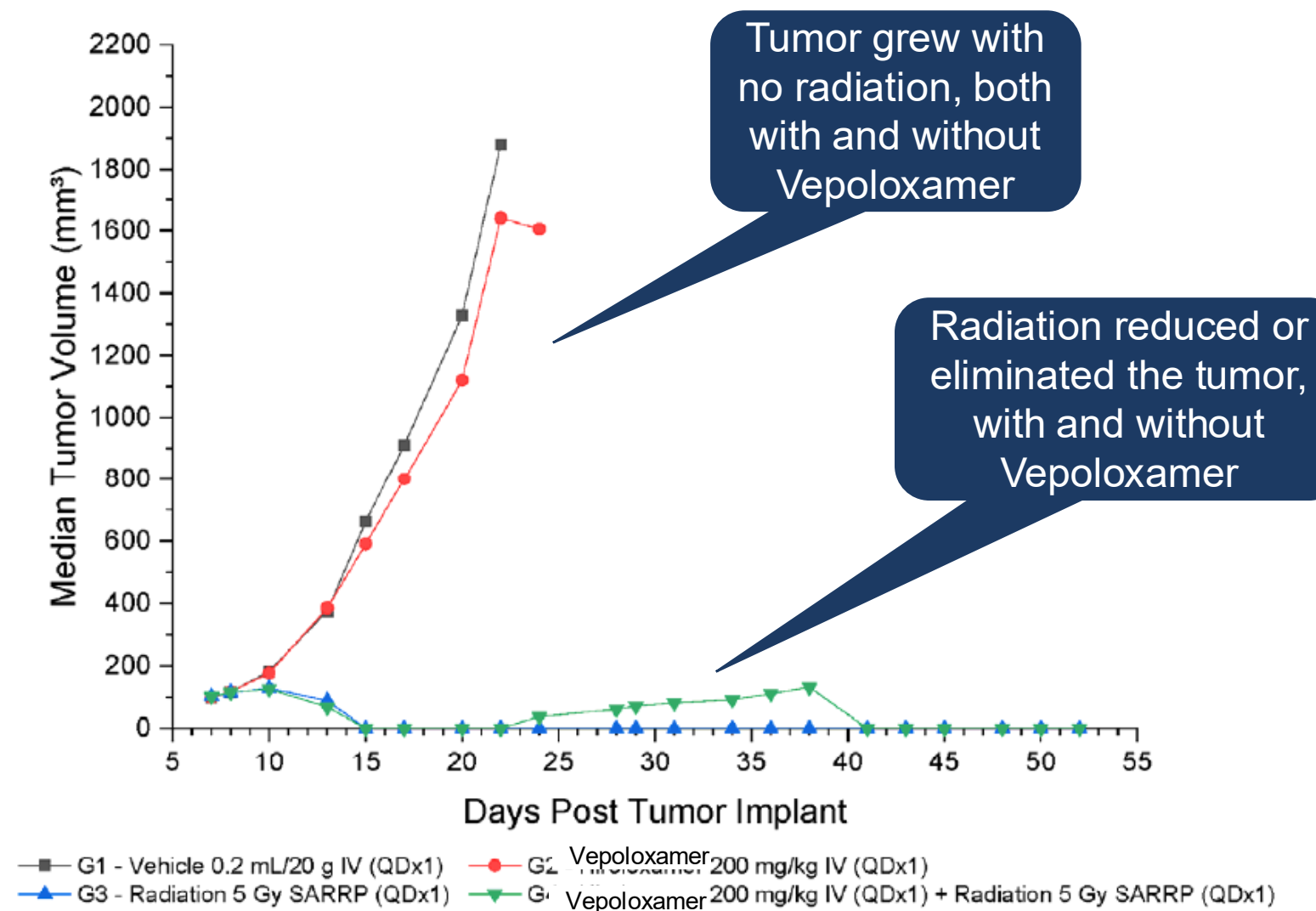
Study Design

- Healthy lung tissue in rats exposed to 20 Gy radiation (targeted to right center lobe of the lung) with and without Vepoloxamer
- Damage (as measured by a thicker alveolar septum) was lessened by the addition of Vepoloxamer prior to radiation

Key findings

- Without Vepoloxamer, radiation caused widespread lung damage that spread far beyond the targeted area
- With Vepoloxamer, the damage was greater in the targeted lobe, and less damage to the tissue that was not targeted

Tumor control: Vepoloxamer does not interfere with radiation efficacy



Study design

- Mouse model with subcutaneous colon carcinoma
- Treated with radiation, with and without Vepoloxamer

Key findings

- Radiation alone shrank or eliminated the tumor in most animals, with the majority still tumor-free at the end of the study
- Adding Vepoloxamer did not blunt that effect

Together with the lung study, these results show the profile of a clinically useful radioprotectant — non-targeted tissue shielded, tumor not.

De-risk in RI-SOM, **build optionality** across the radiation and acute injury franchise



LEAD PROGRAM

De-risk the molecule in severe oral mucositis

- Prophylactic design; on board before the insult
- Lean Phase 1b/2a leveraging the existing IND & data
- AI-enabled drug development; smaller company, faster execution, lower cost



PIPELINE BUILD AND GROWTH OPTIONS

Optionality across radiation & acute injury

- Additional indications where radiation therapy leads to damaged healthy cells
- Acute indications driven by downstream tissue damage
- Non-dilutive funding opportunities: BARDA, NIH/NCI, DoD/CDMRP

PATH TO PHASE 2 READOUT IN RI-SOM

2026

Seed close

Team formation, Pre-IND package, CMC readiness



2027

IND-enabling

GMP DP supply, IND filing



2028

First patient in

Phase 1b/2a in SOM begins



2029

Phase 2a data

Top-line readout → exit / license / grow

Experienced leadership team



John Lunger, MBA

FOUNDING CEO

Former Chief Patient Supply Officer at Adaptimmune (NASDAQ: ADAP); delivered TECELRA, the first FDA-approved engineered T-cell therapy for solid tumors. Built three cell & gene therapy plants and helped raise \$300M+. 25+ years across life sciences in small molecule and biologics, operations, strategy, and corporate development.

Adaptimmune · Wyeth · Pfizer · VWR · Merrimack



Arun Kejariwal, MS, MBA

CHIEF OPERATING OFFICER

Biopharma operations & strategy leader with 20+ years translating scientific insight into high-impact therapies. Portfolio, development and asset-leadership roles spanning gene therapy and CNS.

Spark Therapeutics · GSK · Pfizer



Keith Dewedoff

CHIEF FINANCIAL OFFICER

20+ years as CFO across venture-backed and public biotechs. Concurrent CFO / Advisor at Danforth Advisors and Chair of Kaizen Bioscience. B.S. Economics, Northeastern University.

Danforth Advisors · Kaizen Bioscience

Capital-efficient operating model: CEO full-time; CFO, COO, and additional senior leaders operate as a fractional leadership team as appropriate, with execution delivered through specialist consultants and vendors leveraging AI-native infrastructure

TEAM & ADVISORS

Scientific advisory board

KOLs across the lead indication and the priority pipeline.

RI-SOM · TRANSLATIONAL & CLINICAL EXPERTS

Dr. Stephen Sonis, DMD, DMSc

SAB Chair · Harvard, Brigham & Women's Hospital & Dana-Farber Cancer Institute

Defined highly translational models of cancer-therapy tissue toxicity — the basis of modern oral mucositis drug development. Senior academic at Harvard, Brigham & Women's and Dana-Farber. Founder of Biomodels and Primary Endpoint Solutions. 350+ publications cited 39,000+ times.

Dr. Alessandro Villa, DDS, PhD, MPH

Chief of Oral Medicine & Oral Oncology · Herbert Wertheim Cancer Institute

Leading clinician-scientist in oral complications of cancer therapy — LifeRaft's lead indication. Trained at Harvard/Brigham & Women's and NCI/NIH; former UCSF Chief of Oral Medicine. Specializes in oral mucosal disease, dysplasia, and cancer-treatment toxicities. Subject-matter expert for Primary Endpoint Solutions.

TRAUMA & CRITICAL CARE

Dr. Vanessa Nomellini, MD, PhD, FACS

Associate Professor, Burn/Trauma & Critical Care Surgery · UT Southwestern

Bridges Vepoloxamer into burns, trauma & TBI — the priority pipeline. MD/PhD;

Director of Research for Burn, Trauma & Critical Care Surgery at UT Southwestern.

Holds an NIH R35 on immune suppression in critically ill trauma and sepsis patients.

VEPOLOXAMER · INVENTOR

Dr. Martin Emanuele, PhD, MBA

40+ years biopharma R&D & executive leadership · Vepoloxamer originator

Inventor of Vepoloxamer — led the purified poloxamer program through clinical

development. ~40 years in biopharma R&D and executive leadership. Inventor on 30+

patents; drove 8 first-in-class INDs and 2 NDAs. Closed 20+ partnerships exceeding \$1B in value.

\$30M total raise. \$15M funds through Phase 1b in RI-SOM — the first clinical signal; the next \$15M funds the Phase 2a readout

CATEGORY	ACTIVITY					2027												2028					
		Q3'26	Q4'26			Q1'27			Q2'27			Q3'27			Q4'27			Q1'28			Q2'28		
		Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Apr-27	May-27	Jun-27	Jul-27	Aug-27	Sep-27	Oct-27	Nov-27	Dec-27	Jan-28	Feb-28	Mar-28	Apr-28	May-28	Jun-28
CMC / MFG	API use confirmation testing																						
	CDMO selection · Drug Product for Phase 1b/2																						
	Formulation development and tech transfer																						
	GMP manufacturing and release																						
PRECLINICAL	RI SOM Hamster cheek model																						
REG / IND	IND Gap Analysis and IND planning																						
	IND-enabling lab work																						
	IND filing (prep)																						
	IND Submission																						
CLINICAL	Trial design and execution planning																						
	Trial start-up (CRO, site activation)																						
	Phase 1b — open-label dose escalation																						
	Phase 2a — randomized, double-blind																						
	Phase 1 burns (IIT) — UTSW / Parkland																						

\$1.2M pre-seed completed

\$15M through Phase 1b

+\$15M through Phase 2a

VALUATION INFLECTION POINTS

- ◆ RI-SOM model readout Q4 2026
- ◆ Phase 1b readout early Q3 2028
- ◆ Phase 2a topline mid-2029 · LPI
- ◆ Phase 2a primary CSR Q1 2030

One molecule, many large opportunities

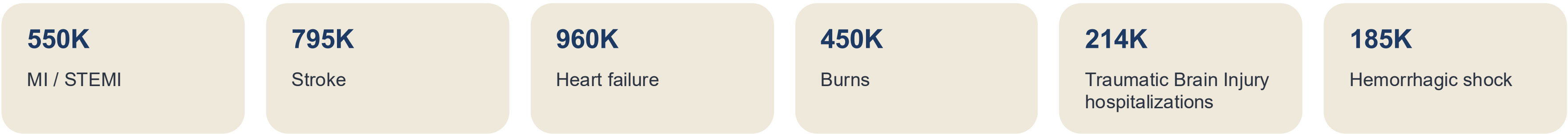
The same membrane-sealing mechanism that protects oral mucosa from radiation injury applies to any condition driven by cell-membrane damage

Scientific validation across multiple indications

60+ independent preclinical studies — peer-reviewed efficacy signals across multiple organ systems. Examples include:



Acute Indications with supporting academic research using P188 or Vepoloxamer — US patients/year



Sources: Preclinical: Wang et al. Stroke 2019; Frontiers Neurology 2023; Chen 2024 Neurochem Int; Sabbah 2020. US incidence: MI/STEMI per StatPearls (550K new MI/yr); Stroke per CDC (795K/yr); Heart failure per HFSA HF Stats 2024 (960K new dx/yr); TBI per CDC (214K hospitalizations, 2020); Burns/Hem. shock per published literature.

Key published research

Peer-reviewed evidence for the Poloxamer 188 / Vepoloxamer membrane-sealing mechanism across indications.

ORAL MUCOSITIS — LEAD INDICATION

Sonis ST. Supportive Care in Cancer, 2021.
<https://doi.org/10.1007/s00520-021-06108-w>

Current Treatment Options in Oncology, 2022.
<https://doi.org/10.1007/s11864-022-00959-z>

Elad S et al. CA: A Cancer Journal for Clinicians, 2022.
<https://doi.org/10.3322/caac.21704>

Nonzee NJ et al. Cancer, 2008. Cost of severe OM.
<https://doi.org/10.1002/cncr.23714>

CARDIAC PROTECTION · HEART FAILURE / MI

Effects of IV Infusion of Vepoloxamer on Left Ventricular Function in Dogs with Advanced Heart Failure

Sabbah HN et al. Cardiovasc Drugs Ther, 2020.
<https://doi.org/10.1007/s10557-020-06953-7>

Cardioprotection by P188 is mediated through increased endothelial nitric oxide production

Chen G et al. Sci Rep, 2025.
<https://doi.org/10.1038/s41598-025-97079-z>

ISCHEMIC STROKE (+ tPA)

Vepoloxamer Enhances Fibrinolysis of tPA on Acute Ischemic Stroke

Wang C et al. Stroke, 2019.
<https://doi.org/10.1161/STROKEAHA.119.026049>

Treatment of stroke in aged male and female rats with Vepoloxamer and tPA reduces neurovascular damage

Zhang ZG et al. Front Neurol, 2023.
<https://doi.org/10.3389/fneur.2023.1282736>

TRAUMATIC BRAIN INJURY

Vepoloxamer improves functional recovery in rat after traumatic brain injury: A dose response and therapeutic window study

Chen L et al. Neurochem Int, 2024.
<https://doi.org/10.1016/j.neuint.2023.105659>

Treatment of TBI with Vepoloxamer, Zhang Y et al. J Neurotrauma, 2018.
<https://doi.org/10.1089/neu.2017.5284>

BURNS

The use of P188 in Burn Injury Treatment; A systemic literature review

Mutore KT et al. Shock, 2024.
<https://doi.org/10.1097/SHK.0000000000002439>

HEMORRHAGIC SHOCK

Autoresuscitation of P188 in Pigs with Traumatic Severe Hemorrhage

Martini WZ et al. Shock, 2022.
<https://doi.org/10.1097/SHK.0000000000001892>